SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3 Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Mohammad Aleyas / 192521220 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mohammad Aleyas

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:
i. D1 cannot work Night shift
ii. D2 must work before D3 (Morning < Afternoon < Night)
iii. Only one doctor per shift
iv. D3 cannot work Morning
v. All doctors must be assigned exactly one shift
Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.
i. Movement allowed: Up, Down, Left, Right
ii. Cost of each move = 1
iii. Cannot pass through obstacles
iv. Use Manhattan Distance heuristic
Using above constraints and BFS Search Algorithm show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.
Constraints:
• The robot can move up, down, left, right
• Some cells are blocked (obstacles) and cannot be traversed
• The robot has limited sensing capability (can only observe adjacent cells)
• Each movement has a cost of 1, but entering risky zones has an additional cost of 2
• The robot must minimize total cost while ensuring safe navigation
• The robot must update its knowledge dynamically (online search scenario)
Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as: Intelligent agents; Rationality; Environment type.
d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWERS

1. Backtracking Search: Doctor–Shift Assignment

Given: Doctors D1, D2, D3; shifts Morning, Afternoon, Night. Constraints: D1 cannot work Night; D2 must be earlier than D3; one doctor per shift; D3 cannot work Morning; each doctor receives exactly one shift.

Step 1 — Assign D2 to Morning. Check: allowed; no other doctor is assigned to Morning.

Step 2 — Try D3 in Afternoon. Check: D2 is before D3, so this satisfies D2 < D3; D3 is not in Morning. Keep this assignment.

Step 3 — Try D1 in Night. Check: invalid, because D1 cannot work Night. Backtrack.

Step 4 — Assign D1 to Afternoon and D3 to Night, with D2 at Morning. Check: D1 is not at Night; D3 is not at Morning; D2 is before D3; every shift has one doctor. All constraints are satisfied.

Final valid schedule: Morning — D2; Afternoon — D1; Night — D3.

Constraint check: D1 ≠ Night ✓; D2 earlier than D3 ✓; one doctor per shift ✓; D3 ≠ Morning ✓; all doctors assigned once ✓.

2. 5×5 Grid: BFS and Manhattan Distance

Note: The question does not provide the exact locations of S, G, or obstacles. For demonstration, assume this grid; # means obstacle:

Row 1: S . # . .
Row 2: . . # . #
Row 3: # . . . .
Row 4: . # # . .
Row 5: . . . . G

Coordinates use (row, column). Start S=(1,1), Goal G=(5,5). Each allowed move costs 1. Manhattan distance is $h(r,c)=|r-5|+|c-5|$.

BFS uses a FIFO queue and expands nodes in increasing depth. Manhattan distance can be recorded as a heuristic, but it does not change BFS's FIFO ordering.

Step-by-step expansion (blocked/out-of-grid cells and previously visited cells are rejected):

1. Start queue [(1,1)], cost 0, h=8. Expand (1,1); enqueue (2,1), (1,2).
2. Expand (2,1); (3,1) is blocked; enqueue (2,2).
3. Expand (1,2); (1,3) is blocked; (2,2) is already visited.
4. Expand (2,2); (2,3) is blocked; enqueue (3,2).
5. Expand (3,2); enqueue (3,3); (4,2) is blocked.
6. Expand (3,3); enqueue (3,4); (4,3) is blocked.
7. Expand (3,4); enqueue (2,4), (4,4), and (3,5) as valid unvisited neighbors.
8. Continue in FIFO order, avoiding obstacles and visited cells, until G=(5,5) is reached.

Trace parent links to obtain a shortest path.

One shortest path for this assumed grid: (1,1) → (2,1) → (2,2) → (3,2) → (3,3) → (3,4) → (4,4) → (5,4) → (5,5).

Path cost = 8 moves × 1 = 8. BFS is optimal because all moves have equal cost. The original question does not show the actual grid, so this is an illustrative example.

3. Autonomous Rescue Robot

(a) Constraint analysis

• Four-direction movement limits actions; diagonal movement is not allowed.
• Obstacles make transitions impossible, so blocked cells must be avoided.
• The robot senses only adjacent cells and cannot know the whole map initially; it must update its knowledge while moving.
• A normal move costs 1. Entering a risky zone adds 2, so that move costs 3 in total.
• Online search requires the robot to sense, update its map, and re-plan as new information appears.

(b) Approach: Uniform Cost Search (UCS) with online sensing
1. Start at S and record the cells currently known.
2. Sense adjacent cells and classify them as free, blocked, or risky when known.
3. Use UCS with cumulative cost g(n): add 1 for a normal move and 3 for entering a risky zone.
4. Select the frontier state with the lowest accumulated cost from the priority queue.
5. Move only through a known traversable cell, then sense adjacent cells again.
6. Update the map and re-run UCS if a newly discovered obstacle or risky cell changes the best route.
7. Stop when G is reached and report the route and total cost. If no known route is available, explore a safe frontier and continue sensing.
Priority queue: insert a state when discovered; update it if a cheaper path is found; always expand the state with the lowest cumulative cost. With non-negative costs, UCS finds the least-cost route in the graph being searched.
(c) AI concepts
• Intelligent agent: the robot senses its surroundings and acts to reach the survivor while avoiding danger.
• Rationality: it chooses the best action based on available information to minimize total cost and navigate safely; it does not assume unseen cells are clear.
• Environment type: partially observable, sequential, uncertain, discrete, and online/dynamic from the robot's perspective.
(d) Justification
UCS compares total path costs and accounts for risky zones, unlike BFS, which assumes equal step costs. Online sensing and re-planning suit a partially observable environment because the robot gradually learns the map and can adapt to new obstacles or hazards.

RUBRICS FOR CALCULATION:

Assessment rubric is provided in the original AT2 handout; refer to that rubric when evaluating this submission.